33. DCC *implies* mC*:* Suppose that mC does not hold in R, and let S be a set of ideals in R where the mC fails, so that every ideal in S does properly contain another ideal of S. Then starting with any ideal $N_1 \in S$, we can find an ideal $N_2 \in S$ properly contained in N_1, and then an ideal N_3 of S properly contained in N_2, etc. This leads to an unending chain of ideals, each properly containing the next, which would contradict the DCC. Thus the DCC implies the mC.

mC *implies* DCC*:* Let $N_1 \supseteq N_2 \supseteq N_3 \supseteq \cdots$ be a descending chain of ideal in R. Let $S = \{N_i \mid i \in \mathbb{Z}^+\}$. By the mC, there is some ideal N_r of S that does not properly contain any other ideal in the set. Thus $N_r = N_{r+1} = N_{r+2} = \cdots$ so the DCC holds.

34. Now $\mathbb{Z}$ is a ring in which the ACC holds, because $\mathbb{Z}$ is PID. However,

$$\mathbb{Z} \supset 2\mathbb{Z} \supset 4\mathbb{Z} \supset \cdots \supset 2^{i-1}\mathbb{Z} \supset \cdots$$

is an infinite descending chain of ideals, so the DCC does not hold in $\mathbb{Z}$.

46. Euclidean Domains

1. Yes, it is a Euclidean norm. To see this, remember that we know $|\ |$ is a Euclidean norm on $\mathbb{Z}$. For Condition 1, find q and r such that $a = bq + r$ where either $r = 0$ or $|r| < |b|$. Then surely we have either $\nu(r) = 0$ or $\nu(r) = r^2 < b^2 = \nu(b)$, because r and b are integers. For Condition 2, note that $\nu(a) = a^2 \leq a^2b^2 = \nu(ab)$ for nonzero a and b, because a and b are integers.

2. No, ν is not a Euclidean norm. Let $a = x$ and $b = 2x$ in $\mathbb{Z}[x]$. There are no $q(x), r(x) \in \mathbb{Z}[x]$ satisfying $x = (2x)q(x) + r(x)$ where the degree of $r(x)$ is less than 1.

3. No, ν is not a Euclidean norm. Let $a = x$ and $b = x + 2$ in $\mathbb{Z}[x]$. There are no $q(x), r(x) \in \mathbb{Z}[x]$ satisfying $x = (x+2)q(x) + r(x)$ where the absolute value of the coefficient of the highest degree term in $r(x)$ is less than 1.

4. No, it is not a Euclidean norm. Let $a = 1/2$ and $b = 1/3$. Then $\nu(a) = (1/2)^2 = 1/4 > 1/36 = \nu(1/6) = \nu(ab)$, so Condition 2 is violated.

5. Yes, it is a Euclidean norm, but not a useful one. Let $a, b, \in \mathbb{Q}$. If $b \neq 0$, let $q = a/b$. Then $a = bq + 0$, which satisfies Condition 1. For Condition 2, if both a and b are nonzero, then $\nu(a) = 50 \leq 50 = \nu(ab)$.

6. We have 23 = 3(138) - 1(391), but 138 = 3,266 - 8(391), so

$$23 = 3[3,266 - 8(391)] - 1(391) = 3(3,266) - 25(391).$$

Now 391 = 7(3,266) - 22,471, so

$$\begin{aligned} 23 &= 3(3,266) - 25[7(3,266) - 22,471] \\ &= 25(22,471) - 172(3,266). \end{aligned}$$

7. Performing the division algorithm, we obtain

$$\begin{aligned} 49,349 &= (15,555)3 + 2,684 \\ 15,555 &= (2,684)6 - 549 \\ 2,684 &= (549)5 - 61 \\ 549 &= (61)9 + 0 \end{aligned}$$

so the gcd is 61.

8. We have 61 = 5(549) - 2,684, but 549 = 6(2,684) - 15,555, so

$$61 = 5[6(2,684) - 15,555] - 2,684 = 29(2,684) - 5(15,555).$$

Now 2,684 = 49,349 - 3(15,555), so

$$\begin{aligned} 61 &= 29[49,349 - 3(15,555)] - 5(15,555) \\ &= 29(49,349) - 92(15,555). \end{aligned}$$

9. We use the division algorithm.

$$\begin{array}{r|l} & x^4 - 2x \\ \hline x^6 - 3x^5 + 3x^4 - 9x^3 + 5x^2 - 5x + 2 & x^{10} - 3x^9 + 3x^8 - 11x^7 + 11x^6 - 11x^5 + 19x^4 - 13x^3 + 8x^2 - 9x + 3 \\ & \underline{x^{10} - 3x^9 + 3x^8 - 9x^7 + 5x^6 - 5x^5 + 2x^4} \\ & \quad -2x^7 + 6x^6 - 6x^5 + 17x^4 - 13x^3 + 8x^2 - 9x + 3 \\ & \quad \underline{-2x^7 + 6x^6 - 6x^5 + 18x^4 - 10x^3 + 10x^2 - 4x} \\ & \quad\quad -x^4 - 3x^3 - 2x^2 - 5x + 3 \end{array}$$

$$\begin{array}{r|l} & -x^2 + 6x - 19 \\ \hline -x^4 - 3x^3 - 2x^2 - 5x + 3 & x^6 - 3x^5 + 3x^4 - 9x^3 + 5x^2 - 5x + 2 \\ & \underline{x^6 + 3x^5 + 2x^4 + 5x^3 - 3x^2} \\ & \quad -6x^5 + x^4 - 14x^3 + 8x^2 - 5x \\ & \quad \underline{-6x^5 - 18x^4 - 12x^3 - 30x^2 + 18x} \\ & \quad\quad 19x^4 - 2x^3 + 38x^2 - 23x + 2 \\ & \quad\quad \underline{19x^4 + 57x^3 + 38x^2 + 95x - 57} \\ & \quad\quad\quad -59x^3 - 118x + 59 \end{array}$$

Multiply by the unit -1/59.

$$\begin{array}{r|l} & -x - 3 \\ \hline x^3 + 2x - 1 & -x^4 - 3x^3 - 2x^2 - 5x + 3 \\ & \underline{-x^4 - 2x^2 + x} \\ & \quad -3x^3 - 6x + 3 \\ & \quad \underline{-3x^3 - 6x + 3} \\ & \quad\quad 0 \end{array}$$

A gcd is $x^3 + 2x - 1$.

10. Use the Euclidean algorithm to find the gcd d_2 of a_2 and a_1. Then use it to find the gcd d_3 of a_3 and d_2. Then use it again to find the gcd d_4 of a_4 and d_3. Continue this process until you find the gcd d_n of a_n and d_{n-1}. The gcd of the n members $a_1, a_2, \cdots, a_n$ is d_n.

11. We use the notation of the solution of the preceding exercise with $a_1 = 2178, a_2 = 396, a_3 = 792$, and $a_4 = 726$. We have 2178 = 5(396) + 198 and 396 = 2(198) + 0, so $d_2 = 198$. We have 792 = 4(198) + 0 so $d_3 = 198$. We have 726 = 3(198) + 132, 198 = 1(132) + 66, and 132 = 2(66) + 0. Thus the gcd of 2178, 396, 792, and 726 is $d_4 = 66$.

12. **a.**Yes, $\mathbb{Z}[x]$ is a UFD because $\mathbb{Z}$ is a UFD and Theorem 45.29 tells us that if D is a UFD, then $D[x]$ is a UFD.

b. The set described consists of all polynomials in $\mathbb{Z}[x]$ that have a constant term in $2\mathbb{Z}$, that is, an even number as constant term. This property of a polynomial is obviously satisfied by $0 \in 2\mathbb{Z}$ and is preserved under addition, subtraction, and multiplication by every element of $\mathbb{Z}[x]$, so the set is an ideal of $\mathbb{Z}[x]$.

c. $\mathbb{Z}[x]$ is not a PID because the ideal described in Part(**b**) is not a principal ideal; a generating polynomial would to have constant term 2, but could not yield, under multiplication by elements of $\mathbb{Z}[x]$, all the polynomials of the form $2 + nx$ in the ideal because n can be odd as well as even.

d. $\mathbb{Z}[x]$ is not an Euclidean domain, because every Euclidean domain is a PID by Theorem 46.4, but Part(**c**) shows that $\mathbb{Z}[x]$ is not a PID.

13. T F T F T T T F T T

14. No, it does not. For any integral domain, its arithmetic structure is completely determined by the binary operations of addition and of multiplication. A Euclidean norm, if one exists, can be used to *study* the arithmetic structure, but it in no way changes it

15. Let a and b be associates of a Euclidean domain D with Euclidean norm ν. Because a and b are associates, there exists a unit u in D such that $a = bu$. Then u^{-1} is also a unit of D and $b = au^{-1}$. Condition 2 of a Euclidean norm yields $\nu(b) \leq \nu(bu) = \nu(a) \leq \nu(au^{-1}) = \nu(b)$. Thus $\nu(a) = \nu(b)$.

16. Suppose that $\nu(a) < \nu(ab)$. If b were a unit, then a and ab would be associates, and by Exercise 15, we would have $\nu(a) = \nu(ab)$. Thus b is not a unit.

For the converse, suppose that $\nu(a) = \nu(ab)$. We claim that then $\langle a\rangle = \langle ab\rangle$, for the proof of Theorem 46.4 shows that a nonzero ideal in a Euclidean domain is generated by any element having minimum norm in the ideal. Thus ab also generates $\langle a\rangle$, so $a = (ab)c$ for some $c \in D$. Then, cancelling the a in the integral domain, we find that $1 = bc$ so b is a unit.

17. The statement is false. For example, let $D = \mathbb{Z}$ and let the norm be $|\ |$. Then $|2| > 1$ and $|-3| > 1$, but $|2 + (-3)| = |-1| = 1$, so the given set is not closed under addition.

18. Let F be a field and let $\nu(a) = 1$ for all $a \in F, a \neq 0$. Because every nonzero element of F is a unit, we see that for nonzero $b \in F$ we have $a = b(a/b) + 0$, which satisfies Condition 1 for a Euclidean norm. Also, $\nu(a) = \nu(ab) = 1$, so Condition 2 is satisfied.

19. **a.** For Condition 1, let $a, b \in D^*$ where $b \neq 0$. There exist $q, r \in D$ such that $a = bq + r$ where either $r = 0$ or $\nu(r) < \nu(b)$. But then either $r = 0$ or $\eta(r) = \nu(r) + s < \nu(b) + s = \eta(b)$, so Condition 1 holds. For $a, b \neq 0$, we have $\eta(a) = \nu(a) + s \leq \nu(ab) + s = \eta(ab)$, so Condition 2 holds. The hypothesis that $\nu(1) + s > 0$ guarantees that $\eta(a) > 0$ for all $a \in D^*$, because $\nu(1)$ is minimal among all $\nu(a)$ for $a \in D^*$, by Theorem 46.6.

b. For Condition 1, let $a, b \in D^*$ where $b \neq 0$. There exist $q, r \in D$ such that $a = bq + r$ where either $r = 0$ or $\nu(r) < \nu(b)$. But then either $r = 0$ or $\lambda(r) = t \cdot \nu(r) < t \cdot \nu(b) = \lambda(b)$, so Condition 1 holds. For $a, b \neq 0$, we have $\lambda(a) = t \cdot \nu(a) \leq t \cdot \nu(ab) = \lambda(ab)$, so Condition 2 holds. The hypothesis that $t \in \mathbb{Z}^+$ guarantees that $\lambda(a) > 0$ and $\lambda(a)$ is an integer for all $a \in D^*$.

c. Let ν be a Euclidean norm on D. Then $\lambda(a) = 100 \cdot \nu(a)$ for $a \in D^*$ is a Euclidean norm on D by Part(**b**). Let $s = 1 - \lambda(1)$. Then $\mu(a) = \lambda(a) + s$ for $a \in D^*$ is a Euclidean norm on D by Part(**a**), and $\mu(1) = \lambda(1) + s = \lambda(1) + [1 - \lambda(1)] = 1$. If $a \neq 0$ is a nonunit in D, then $\nu(a) \geq \nu(1) + 1$ so $\lambda(a) = 100 \cdot \nu(a) \geq 100 \cdot [\nu(1) + 1]$ and $\mu(a) = \lambda(a) + s \geq 100 \cdot [\nu(1) + 1] + 1 - \lambda(1) = 100 \cdot [\nu(1) + 1] + 1 - 100 \cdot \nu(1) = 100 + 1 = 101$.

20. We know that all multiples of $a \in D$ form the principal ideal $\langle a\rangle$ and all multiples of b form the principal ideal $\langle b\rangle$. By Exercise 27 of Section 26, the intersection of ideals in a ring is an ideal, so $\langle a\rangle \cap \langle b\rangle$ is an ideal, and consists of all common multiples of a and b. Because a Euclidean domain is a PID, this ideal has a generator c. Now $a \mid c$ and $b \mid c$ because c is a common multiple of a and b.